UG (NEP) 7th SEMESTER
COMPUTER APPLICATIONS: MACHINE LEARNING
(CAP722J3)

Session: February 2026 Time Allowed: 2:00 Hours
Max. Marks: 100 Min. Marks: 36

NOTE: Attempt all questions from Section “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

I. What is Machine Learning?

II. Define supervised learning with an example.

III. What is feature extraction?

IV. What is the main difference between training and testing data?

V. Define clustering in machine learning.

VI. What is Naïve Bayes Classifier used for?

VII. What is the role of a kernel function in SVM?

VIII. Mention one application of PCA in data analysis.

Section B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all the four questions with internal choice in a maximum of 150 words each.

1. Explain types of learning in Machine Learning.

OR

Explain the main challenges of Machine Learning.

2. Describe the working of Decision Tree Learning Algorithm.

OR

Discuss issues in Decision Tree Learning.

3. Explain K-Means clustering algorithm with an example.

OR

Explain Hierarchical clustering method.

4. Describe Bayesian Learning and its applications.

OR

Explain the concept of Support Vector Machine (SVM).

Section C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

1. Explain in detail the process of designing a learning system. Discuss the importance of testing and validation in Machine Learning.
2. Describe different clustering algorithms (K-Means, DBSCAN, Fuzzy C-Means, and Hierarchical Clustering) with their advantages and limitations.
3. Explain the working of Linear and Logistic Regression with suitable examples. Discuss how performance is evaluated using confusion matrix metrics.
4. Discuss Support Vector Machines in detail. Explain how kernel tricks help in solving non-linear classification problems.